Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Abstract

This project analyzes profitability using sales, cost, gross profit, and operational data to identify profitable products, compare divisions, evaluate margins, and support business decision making. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

This project analyzes profitability using sales, cost, gross profit, and operational data to identify profitable products, compare divisions, evaluate margins, and support business decision making. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

This project analyzes profitability using sales, cost, gross profit, and operational data to identify profitable products, compare divisions, evaluate margins, and support business decision making. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

This project analyzes profitability using sales, cost, gross profit, and operational data to identify profitable products, compare divisions, evaluate margins, and support business decision making. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

This project analyzes profitability using sales, cost, gross profit, and operational data to identify profitable products, compare divisions, evaluate margins, and support business decision making. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

This project analyzes profitability using sales, cost, gross profit, and operational data to identify profitable products, compare divisions, evaluate margins, and support business decision making. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Introduction

Nassau Candy Distributor requires insight into which products generate sustainable profits. The project combines exploratory data analysis, feature engineering, machine learning, and dashboard development. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Nassau Candy Distributor requires insight into which products generate sustainable profits. The project combines exploratory data analysis, feature engineering, machine learning, and dashboard development. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Nassau Candy Distributor requires insight into which products generate sustainable profits. The project combines exploratory data analysis, feature engineering, machine learning, and dashboard development. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Nassau Candy Distributor requires insight into which products generate sustainable profits. The project combines exploratory data analysis, feature engineering, machine learning, and dashboard development. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Nassau Candy Distributor requires insight into which products generate sustainable profits. The project combines exploratory data analysis, feature engineering, machine learning, and dashboard

development. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Nassau Candy Distributor requires insight into which products generate sustainable profits. The project combines exploratory data analysis, feature engineering, machine learning, and dashboard development. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Problem Statement

The organization lacks clear visibility into profitability across products and divisions, making pricing and portfolio decisions difficult. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The organization lacks clear visibility into profitability across products and divisions, making pricing and portfolio decisions difficult. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The organization lacks clear visibility into profitability across products and divisions, making pricing and portfolio decisions difficult. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The organization lacks clear visibility into profitability across products and divisions, making pricing and portfolio decisions difficult. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The organization lacks clear visibility into profitability across products and divisions, making pricing and portfolio decisions difficult. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The organization lacks clear visibility into profitability across products and divisions, making pricing and portfolio decisions difficult. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Objectives

Calculate profitability metrics, identify profitable products, compare divisions, perform Pareto analysis, develop machine learning models, and build a Streamlit dashboard. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Calculate profitability metrics, identify profitable products, compare divisions, perform Pareto analysis, develop machine learning models, and build a Streamlit dashboard. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Calculate profitability metrics, identify profitable products, compare divisions, perform Pareto analysis, develop machine learning models, and build a Streamlit dashboard. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Calculate profitability metrics, identify profitable products, compare divisions, perform Pareto analysis, develop machine learning models, and build a Streamlit dashboard. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Calculate profitability metrics, identify profitable products, compare divisions, perform Pareto analysis, develop machine learning models, and build a Streamlit dashboard. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Calculate profitability metrics, identify profitable products, compare divisions, perform Pareto analysis, develop machine learning models, and build a Streamlit dashboard. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dataset Description

The dataset includes order details, customer information, product divisions, regions, sales, units, costs, and gross profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The dataset includes order details, customer information, product divisions, regions, sales, units, costs, and gross profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The dataset includes order details, customer information, product divisions, regions, sales, units, costs, and gross profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The dataset includes order details, customer information, product divisions, regions, sales, units, costs, and gross profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The dataset includes order details, customer information, product divisions, regions, sales, units, costs, and gross profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The dataset includes order details, customer information, product divisions, regions, sales, units, costs, and gross profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Methodology

Data cleaning, feature engineering, exploratory analysis, visualization, predictive modeling, and dashboard development were performed. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Data cleaning, feature engineering, exploratory analysis, visualization, predictive modeling, and dashboard development were performed. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Data cleaning, feature engineering, exploratory analysis, visualization, predictive modeling, and dashboard development were performed. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Data cleaning, feature engineering, exploratory analysis, visualization, predictive modeling, and dashboard development were performed. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Data cleaning, feature engineering, exploratory analysis, visualization, predictive modeling, and dashboard development were performed. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Data cleaning, feature engineering, exploratory analysis, visualization, predictive modeling, and dashboard development were performed. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Data Cleaning and Validation

Dates were standardized, missing values handled, invalid records checked, and numerical fields validated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dates were standardized, missing values handled, invalid records checked, and numerical fields validated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dates were standardized, missing values handled, invalid records checked, and numerical fields validated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dates were standardized, missing values handled, invalid records checked, and numerical fields validated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dates were standardized, missing values handled, invalid records checked, and numerical fields validated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dates were standardized, missing values handled, invalid records checked, and numerical fields validated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Feature Engineering

Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Month fields were created. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Month fields were created. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Month fields were created. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Month fields were created. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Month fields were created. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Gross Margin (%), Profit per Unit, Revenue Contribution, Profit Contribution, and Month fields were created. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Exploratory Data Analysis

Product, division, regional, monthly, and Pareto analyses were conducted using Plotly visualizations. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Product, division, regional, monthly, and Pareto analyses were conducted using Plotly visualizations. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Product, division, regional, monthly, and Pareto analyses were conducted using Plotly visualizations. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Product, division, regional, monthly, and Pareto analyses were conducted using Plotly visualizations. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Product, division, regional, monthly, and Pareto analyses were conducted using Plotly visualizations. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Product, division, regional, monthly, and Pareto analyses were conducted using Plotly visualizations. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Machine Learning

Linear Regression and Random Forest Regression were implemented and evaluated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Linear Regression and Random Forest Regression were implemented and evaluated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Linear Regression and Random Forest Regression were implemented and evaluated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Linear Regression and Random Forest Regression were implemented and evaluated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Linear Regression and Random Forest Regression were implemented and evaluated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Linear Regression and Random Forest Regression were implemented and evaluated. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Dashboard

The Streamlit dashboard includes KPI cards, filters, profitability charts, Pareto analysis, trend analysis, and downloadable filtered data. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The Streamlit dashboard includes KPI cards, filters, profitability charts, Pareto analysis, trend analysis, and downloadable filtered data. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The Streamlit dashboard includes KPI cards, filters, profitability charts, Pareto analysis, trend analysis, and downloadable filtered data. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The Streamlit dashboard includes KPI cards, filters, profitability charts, Pareto analysis, trend analysis, and downloadable filtered data. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The Streamlit dashboard includes KPI cards, filters, profitability charts, Pareto analysis, trend analysis, and downloadable filtered data. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The Streamlit dashboard includes KPI cards, filters, profitability charts, Pareto analysis, trend analysis, and downloadable filtered data. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Business Insights

Chocolate products generated strong profitability, while Pareto analysis showed that a relatively small number of products contributed most of the profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Chocolate products generated strong profitability, while Pareto analysis showed that a relatively small number of products contributed most of the profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Chocolate products generated strong profitability, while Pareto analysis showed that a relatively small number of products contributed most of the profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Chocolate products generated strong profitability, while Pareto analysis showed that a relatively small number of products contributed most of the profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Chocolate products generated strong profitability, while Pareto analysis showed that a relatively small number of products contributed most of the profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Chocolate products generated strong profitability, while Pareto analysis showed that a relatively small number of products contributed most of the profit. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Recommendations

Prioritize high-margin products, optimize pricing, reduce production costs, and monitor KPIs continuously. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Prioritize high-margin products, optimize pricing, reduce production costs, and monitor KPIs continuously. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Prioritize high-margin products, optimize pricing, reduce production costs, and monitor KPIs continuously. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Prioritize high-margin products, optimize pricing, reduce production costs, and monitor KPIs continuously. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Prioritize high-margin products, optimize pricing, reduce production costs, and monitor KPIs continuously. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Prioritize high-margin products, optimize pricing, reduce production costs, and monitor KPIs continuously. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

Conclusion

The solution provides data-driven support for strategic decisions and improves understanding of profitability. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The solution provides data-driven support for strategic decisions and improves understanding of profitability. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The solution provides data-driven support for strategic decisions and improves understanding of profitability. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The solution provides data-driven support for strategic decisions and improves understanding of profitability. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The solution provides data-driven support for strategic decisions and improves understanding of profitability. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

The solution provides data-driven support for strategic decisions and improves understanding of profitability. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.

References

Python for Data Analysis; Pandas Documentation; Plotly Documentation; Streamlit Documentation; Scikit-learn Documentation. The findings support informed decision-making, improve operational efficiency, and demonstrate practical applications of data analytics in business performance management.